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Functional equations at BMO2 and IMO level

Level 5 Advanced · Algebra

Functional equations with no regularity handed to you: Cauchy with a boundedness condition, where that condition hides in an equation, involutions and the pointwise trap, fixed points, equations on the integers, and complete write-ups.

By the end you can: You will be able to prove that a bounded or monotone additive function is linear, and find the hidden regularity in an equation over the reals; You will be able to extract injectivity, surjectivity, involutions and shifts, and rule out pointwise mixtures of solutions; You will be able to solve equations on the integers using shifts, positivity and injectivity, and write a complete 'find all' solution.

Lesson

Find all functions f from the positive reals to the positive reals such that

$$f\big(x+f(y)\big)=f(x)+2y \quad \text{for all } x,y>0.$$

If $f(a)=f(b)$, comparing the equation at (x,a) and (x,b) gives $2a=2b$, so f is injective. Applying the equation twice,

$$f\big(x+f(y)+f(z)\big)=f\big(x+f(y)\big)+2z=f(x)+2y+2z=f\big(x+f(y+z)\big),$$

and injectivity gives $f(y)+f(z)=f(y+z)$. So f is additive on the positive reals. Since f takes positive values, $f(y+z)>f(y)$, so f is increasing. Additivity gives $f(q)=kq$ for positive rationals q, where $k=f(1)$, and for real x and rationals $p<x<q$ we get $kp<f(x)<kq$; letting $p,q\to x$ gives $f(x)=kx$. Substituting, $k^2=2$, so $f(x)=\sqrt{2}\,x$, which works.

Now change the domain and codomain to $\mathbb{R}$. The same argument shows f is additive with $f(f(y))=2y$, and every such function is a solution. But the step "positive values, so increasing" is gone, and there are additive functions f with $f(f(y))=2y$ for all y that are not of the form kx (they are built from a basis of $\mathbb{R}$ over $\mathbb{Q}$, using the axiom of choice). So over $\mathbb{R}$ the answer is not $\pm\sqrt{2}\,x$; it is a set of functions you cannot write down. Problem setters at this level avoid such equations, so an olympiad equation over $\mathbb{R}$ almost always contains something that forces regularity. Finding that something is the main subject of this unit. We assume L4-AL1 (substitutions, injectivity, Cauchy over $\mathbb{Q}$).

Additive functions and regularity

Call $f:\mathbb{R}\to\mathbb{R}$ additive if $f(x+y)=f(x)+f(y)$ for all x,y. From L4-AL1, an additive f satisfies $f(qx)=qf(x)$ for all rational q and real x.

Theorem 1. Let f be additive and bounded above (or bounded below) on some interval (a,b). Then $f(x)=cx$ for all x, where $c=f(1)$.
Proof. Replacing f by $-f$ if necessary, suppose $f\le M$ on (a,b). Let $g(x)=f(x)-cx$. Then g is additive, $g(q)=0$ for rational q, and $g\le M'=M+|c|(|a|+|b|)$ on (a,b). Fix a real x and a positive integer n, and choose a rational q with $nx+q\in(a,b)$. Then $ng(x)=g(nx+q)\le M'$. The same with $-x$ gives $-ng(x)\le M'$. Hence $|g(x)|\le M'/n$ for every n, so $g(x)=0$.

In particular a monotone additive function is linear, and so is an additive function that is non-negative on $[0,\infty)$. In an olympiad equation, the regularity comes from one of the following.

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1. A positive codomain, as in the opening problem.
2. Squares: a relation such as $f(x^2) = f(x)^2$ gives $f \geq 0$ on $[0, \infty)$.
3. A polynomial identity. If f is additive and satisfies an identity involving products like $xf(x)$, replace x by $x + q$ with q rational. Both sides become polynomials in q, and comparing coefficients gives new relations, often linear ones (Polynomial comparison).

Example 1. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x^3 + y) = x^2 f(x) + f(y)$ for all real x, y.

Solution.
Every $f(x) = kx$ works, since $k(x^3+y) = x^2 \cdot kx + ky$. We show there are no others. Let $c = f(0)$. Putting $y = 0$ gives $f(x^3) = x^2 f(x) + c$, so the equation reads $f(x^3 + y) = f(x^3) + f(y) - c$. Every real number is a cube, so $g = f - c$ is additive: $g(t + y) = g(t) + g(y)$ for all t, y. Then $f(x^3) = x^2 f(x) + c$ becomes $g(x^3) = x^2 g(x) + cx^2$, and $x = 1$ gives $c = 0$. So f is additive and $f(x^3) = x^2 f(x)$.
Additivity alone allows wild solutions, so the identity must be used. Replace x by $x + 1$ and expand both sides using additivity, with $a = f(1)$:

$$f(x^3) + 3f(x^2) + 3f(x) + a = (x + 1)^2 \big(f(x) + a\big).$$

Cancelling $f(x^3) = x^2 f(x)$ leaves

$$3f(x^2) + 2f(x) = 2xf(x) + (x^2 + 2x)a.$$

An unknown $f(x^2)$ remains, so we need a second relation; replacing x by $-x$ (note f is odd) gives $3f(x^2) - 2f(x) = 2xf(x) + (x^2 - 2x)a$. Subtracting, $4f(x) = 4ax$. Hence $f(x) = ax$, and the solutions are $f(x) = kx$ for every real k.
No inequality was needed: the term $x^2 f(x)$ mixes f with multiplication by x, and a wild additive function cannot satisfy such an identity.

Injectivity, involutions and the pointwise trap

The first substitutions at this level are the same as in L4-AL1, but the aim is different: before computing anything, extract structure. A relation $f(f(y)) = y + c$ gives a bijection; if $c = 0$, f is an involution, and you may replace y by $f(y)$ anywhere. A zero $f(z) = 0$ (which exists if f is surjective) is usually worth substituting. If $f(f(x)) = x$, every value of f is matched: f swaps x and $f(x)$, and the fixed points of f are the x with $f(x) = x$.

The trap is the same as at Level 4, with harder equations. From $f(x)^2 = x^2$ for all x you know only that for each x separately, $f(x) = x$ or $f(x) = -x$. To finish, assume $f(a) = -a$ and $f(b) = b$ for some nonzero a, b, and derive a contradiction or the structure of all such a, b.

Example 2. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f\big(f(x)^2 + f(y)\big) = xf(x) + y$ for all real x, y.

Solution.
Both $f(x) = x$ and $f(x) = -x$ work: for $f(x) = x$ both sides equal $x^2 + y$, and for $f(x) = -x$ the left side is $f(x^2 - y) = y - x^2$, which is the right side. Write $P(x, y)$ for the equation. $P(0, y)$ gives $f\big(f(0)^2 + f(y)\big) = y$, so f is a bijection. Let z be the number with $f(z) = 0$. Then $P(z, y)$ gives $f(f(y)) = y$, so $P(0, y)$ reads $f\big(f(0)^2 + f(y)\big) = f(f(y))$, and injectivity gives $f(0) = 0$.
$P(x, 0)$ gives $f\big(f(x)^2\big) = xf(x)$. Applying this with x replaced by $f(x)$, and using $f(f(x)) = x$, gives $f(x^2) = f(x)\, x$. Hence $f\big(f(x)^2\big) = f(x^2)$, and injectivity gives $f(x)^2 = x^2$ for every x.

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Now the pointwise trap. Suppose $f(a) = -a$ for some $a \neq 0$. Then $P(a, f(t))$ reads $f\big(a^2 + f(f(t))\big) = -a^2 + f(t)$, that is,

$$f(t + a^2) = f(t) - a^2 \quad \text{for all } t.$$

Squaring and using $f(s)^2 = s^2$ at $s = t + a^2$ gives $(t + a^2)^2 = \big(f(t) - a^2\big)^2$. If $f(t) = t$, this reads $(t + a^2)^2 = (t - a^2)^2$, so $4a^2t = 0$ and $t = 0$. Hence $f(t) = -t$ for every t. So either $f(x) = x$ for all x, or there is some $a \neq 0$ with $f(a) = -a$ and then $f(x) = -x$ for all x. The solutions are $f(x) = x$ and $f(x) = -x$.

Equations on $\mathbb{N}$ and $\mathbb{Z}$

On the integers there is no regularity issue; Cauchy's equation on $\mathbb{Z}$ gives $f(n) = nf(1)$ by induction. The difficulties are different. You cannot put $x = 0$ if 0 is not in the domain, you cannot divide, and the conclusion often depends on positivity of values. Standard tools: show a shift $f(m + d) = f(m) + e$, then compare shifts by different amounts (two shifts commute); use that a positive integer sequence cannot decrease forever; use induction on n.

Example 3. Find all functions f from the positive integers to the positive integers such that $f\big(m + f(n)\big) = f\big(f(m)\big) + n$ for all positive integers m, n.

Solution.

Trying $f(n) = n + e$ gives $m + n + 2e$ on both sides, so every such function with $e \geq 0$ works. We show there are no others. If $f(n_1) = f(n_2)$, comparing $P(1, n_1)$ and $P(1, n_2)$ gives $n_1 = n_2$, so f is injective. Let $a = f(1)$. $P(m, 1)$ gives $f(f(m)) = f(m + a) - 1$, and the equation becomes

$$f\big(m + f(n)\big) = f(m + a) + n - 1. \qquad (*)$$

With $n = 2$ and $b = f(2)$, $(*)$ gives $f(m + b) = f(m + a) + 1$ for all $m \geq 1$, and $b \neq a$.

If $b < a$, put $D = a - b$: then $f(s + D) = f(s) - 1$ for all $s \geq b + 1$, so $f(s + kD) = f(s) - k$, which is negative for large k. This is impossible, so $b > a$, and with $D = b - a$ we get $f(s + kD) = f(s) + k$ for all $s \geq a + 1$ and $k \geq 0$ (Two shifts). Hence $f(m + a) + n - 1 = f\big(m + a + (n - 1)D\big)$, and $(*)$ with injectivity gives

$$f(n) = a + (n - 1)D \quad \text{for all } n.$$

Substituting $f(n) = Dn + e$ with $e = a - D$ into the equation gives $Dm + D^2n + De + e = D^2m + De + e + n$, so $D = D^2$ and $D^2 = 1$, hence $D = 1$. So $f(n) = n + e$ with $e = a - 1 \geq 0$, and we checked above that all of these work.

Positivity was used once, to rule out $b < a$. Over $\mathbb{Z}$ that case would need a different argument.

Fixed points

If the equation produces many points with $f(t) = t$, find what operations the set of fixed points is closed under (Fixed point method). The standard instance is $f : \mathbb{R}^+ \to \mathbb{R}^+$ with $f\big(xf(y)\big) = yf(x)$, the equation of IMO 1983 Problem 1 (where it comes with the extra condition that $f(x) \to 0$ as $x \to \infty$). Putting $x = y = 1$ gives $f(f(1)) = f(1)$; putting $x = 1$ gives $f(f(y)) = f(1)y$, so f is injective, and then $f(f(1)) = f(1)$ gives $f(1) = 1$ and $f(f(y)) = y$. Putting $y = x$ gives $f\big(xf(x)\big) = xf(x)$, so every number $xf(x)$ is a fixed point. If a, b are fixed points, then $f(ab) = f\big(af(b)\big) = bf(a) = ab$, and $1 = f(1) = f\big(\frac{1}{a} \cdot a\big) = f\big(\frac{1}{a} f(a)\big) = af\big(\frac{1}{a}\big)$, so $\frac{1}{a}$ is fixed. Hence if some fixed point $a \neq 1$ exists, all powers a^n $(n \in \mathbb{Z})$ are fixed points, and the set of fixed points is unbounded above. Whatever extra condition the problem gives (a limit, a bound, monotonicity) is then used against this unbounded set; in IMO 1983 it shows that 1 is the only fixed point, so

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$xf(x) = 1$.

Note the structure: the equation alone allowed both $f(x) = x$ and $f(x) = \frac{1}{x}$ (and wild multiplicative involutions), and the extra condition chose between them through the fixed points.

The other Cauchy equations

Three relatives of Cauchy's equation reduce to it, and Theorem 1 transfers through the reduction.

1. $f : \mathbb{R}^+ \to \mathbb{R}$ with $f(xy) = f(x) + f(y)$: put $g(t) = f(e^t)$; then g is additive on $\mathbb{R}$.
2. $f : \mathbb{R} \to \mathbb{R}^+$ with $f(x + y) = f(x)f(y)$: put $g = \ln f$; then g is additive. If f may take the value 0, note that $f(x_0) = 0$ gives $f(x) = f(x_0)f(x - x_0) = 0$ for all x.
3. $f : \mathbb{R}^+ \to \mathbb{R}^+$ with $f(xy) = f(x)f(y)$: put $g(t) = \ln f(e^t)$.

In each case a bound on f over an interval becomes a bound on g over an interval, so the regular solutions are $c \ln x$, e^{cx} and x^c respectively. On $\mathbb{R}$ rather than $\mathbb{R}^+$, the multiplicative equation also needs the sign of $f(-1)$ and the value $f(0)$, which you must find separately.

Writing a complete "find all"

A full-marks solution to a functional equation has the following shape.

1. State the answer at the start, and verify each solution for all values of the variables. Write the check out; "it is easy to verify" loses marks when the check is not easy.
2. Name the substitutions: $P(x, y)$ for the equation, then "$P(0, y)$ gives ...".
3. Prove every structural claim (injective, surjective, odd, additive) before using it, and never divide by $f(x)$ or x without dealing with the zero case.
4. If additivity appears over $\mathbb{R}$, say exactly which regularity condition you then prove, and prove it.
5. Rule out mixtures explicitly.

Common traps.

- Quoting "Cauchy's equation, so $f(x) = cx$" over $\mathbb{R}$ with no regularity. This is wrong, and markers treat it as a serious gap.
- The pointwise trap: $f(x) \in \{x, -x\}$ for each x is not the same as $f(x) = x$ for all x or $f(x) = -x$ for all x.
- Using a value outside the domain, such as $m = 0$ on the positive integers.
- Claiming surjectivity from an expression that covers only part of the line, such as $f(f(y)) = y^2 + c$.
- Checking the answer at a few values instead of for all values.

How to spot this in a contest

BMO Round 2 often sets a functional equation (in 2009, 2019, 2022 and 2024, for example), frequently on the integers or the positive reals, and the IMO has set one in most years since 2008, usually over $\mathbb{R}$ or $\mathbb{R}^+$. In practice:

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1. Guess the answers first (linear, constant, x^2, $1/x$) and check them; the answer tells you which facts to aim for.
2. Look for an $f(f(\cdot))$ term or a variable appearing alone on one side: these give injectivity or surjectivity.
3. When additivity appears, look at the equation for what can replace continuity: a positive codomain, a square, or a product with x.
4. On $\mathbb{N}$ and $\mathbb{Z}$, look for a shift, then use positivity, injectivity and induction.
5. Before writing up, ask whether any step was only pointwise.

Practice

Try each problem before turning to the hints and solutions at the back.

1. ⋆⋆ Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that $f\big(f(m) + n\big) = m + f(n + 2)$ for all integers m, n.
2. ⋆⋆ Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f\big(xf(y) + f(x)\big) = xy$ for all real x, y.
3. ⋆⋆⋆ Prove that there is no function $f : \mathbb{R} \to \mathbb{R}$ such that $f\big(x + f(y)\big) = f(x) + y^2$ for all real x, y.
4. ⋆⋆⋆ Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f\big(x^3 + f(y)\big) = xf(x^2) + y$ for all real x, y.
5. ⋆⋆⋆⋆ Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x^3 + y) = f(x)^3 + f(y)$ for all real x, y.
6. ⋆⋆⋆⋆ Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x^2 + y) + f(x) = f(x)f(x + 1) + f(y)$ for all real x, y.

Contest ladder

Real past problems that use this idea, easiest first. Open them on the official paper or the AoPS wiki, or online at kevinvariant08.github.io.

1. BMO2 2023/24 Problem 2 ⋆⋆
2. BMO2 2008/09 Problem 3 ⋆⋆⋆
3. BMO2 2018/19 Problem 4 ⋆⋆⋆
4. BMO2 2021/22 Problem 2 ⋆⋆⋆
5. IMO 2010 Problem 1 ⋆⋆⋆
6. IMO 2019 Problem 1 ⋆⋆⋆
7. IMO 2008 Problem 4 ⋆⋆⋆
8. IMO 2015 Problem 5 ⋆⋆⋆⋆
9. USAMO 2016 Problem 4 ⋆⋆⋆⋆
10. IMO 2017 Problem 2 ⋆⋆⋆⋆

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Hints

1. Show that f is surjective, then substitute an m with $f(m) = 1$.
2. Find $f(0)$ using the zero of f, then replace y by $f(z)$ and choose z well.
3. Show that f takes every value in $[f(0), \infty)$, and use this to show that $f - f(0)$ is additive.
4. Show that f is an involution and, using that every real number is a cube, that f is additive. Then substitute $x \mapsto x + 1$ and $x \mapsto -x$ in the relation for $f(x^3)$.
5. Show that f is additive. Then expand $f\big((x+q)^3\big) = f(x+q)^3$ for rational q and compare coefficients of powers of q.
6. Show that $f - f(0)$ is additive. Then compare the equation at x and at $-x$, and do not forget constant solutions.

Solutions

Problem 1

The solutions are $f(n) = n+2$ and $f(n) = 2-n$. Check: for $f(n) = n+2$ both sides equal $m+n+4$; for $f(n) = 2-n$ the left side is $f(2-m+n) = m-n$ and the right side is $m + \big(2-(n+2)\big) = m-n$.

Conversely, putting $n = 0$ gives $f(f(m)) = m + f(2)$, and the right side takes every integer value as m varies, so f is surjective. Choose u with $f(u) = 1$. Putting $m = u$ gives $f(n+1) = u + f(n+2)$ for all n, that is, $f(k+1) - f(k) = -u$ for every integer k. Hence $f(n) = b + dn$ with $b = f(0)$ and $d = -u$.

Substituting into the equation: the left side is $b + d(b + dm + n) = d^2m + dn + db + b$ and the right side is $m + b + dn + 2d$. Comparing coefficients of m gives $d^2 = 1$, and comparing constants gives $db = 2d$, so $b = 2$. Hence $f(n) = n + 2$ or $f(n) = 2 - n$, and both were checked above.

Problem 2

The only solution is $f(x) = 1-x$. Check: $xf(y)+f(x) = x-xy+1-x = 1-xy$, and $f(1-xy) = xy$.

Write $P(x, y)$ for the equation. If $f(a) = f(b)$, then $P(1, a)$ and $P(1, b)$ give $a = b$, so f is injective; $P(1, y)$ shows f takes every real value y, so f is surjective. Let $c = f(0)$. $P(0, y)$ gives $f(c) = 0$. $P(x, c)$ gives $f(f(x)) = cx$, and then $x = c$ gives $f(0) = c^2$, so $c = c^2$.

If $c = 0$, then $f(f(x)) = 0 = f(f(0))$ for all x, and injectivity (applied twice) gives $x = 0$ for all x, which is absurd. Hence $c = 1$: $f(0) = 1$, $f(1) = 0$ and $f(f(x)) = x$.

Now $P(x, f(z))$ gives $f\big(xz + f(x)\big) = xf(z)$, and with $z = 1$ this is $f\big(x + f(x)\big) = 0 = f(1)$. By injectivity $x + f(x) = 1$, so $f(x) = 1 - x$ for every x.

Problem 3

Suppose such an f exists and let $c = f(0)$. Putting $x = 0$ gives $f(f(y)) = c + y^2$, so f takes every value in $[c, \infty)$. Let S be the set of values of f, so $[c, \infty) \subseteq S$.

For $t \in S$, say $t = f(y)$, the equation reads $f(x + t) = f(x) + y^2$, and $y^2 = f(f(y)) - c = f(t) - c$. Hence, with $g = f - c$,

$$g(x + t) = g(x) + g(t) \quad \text{for all real } x \text{ and all } t \in S.$$

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We show that g is additive. Given real x, u, choose $T \geq \max(c, c-u)$, so that $T \in S$ and $u+T \in S$. Then

$$g(x+u)+g(T) = g(x+u+T) = g(x)+g(u+T) = g(x)+g(u)+g(T),$$

using the displayed rule with $t = T$, then $t = u+T$, then $t = T$ again (with x replaced by u). Hence $g(x+u) = g(x)+g(u)$, and in particular $g(0) = 0$.

Now $f(f(y)) = c+y^2$ becomes $g\big(g(y)+c\big)+c = c+y^2$, that is, $g(g(y))+g(c) = y^2$. With $y = 0$ this gives $g(c) = 0$, so $g(g(y)) = y^2$. But $g \circ g$ is additive, so $g(g(2)) = 2g(g(1))$, which says $4 = 2$. This contradiction shows that no such f exists.

Problem 4

The solutions are $f(x) = x$ and $f(x) = -x$. Check: for $f(x) = x$ both sides equal x^3+y; for $f(x) = -x$ the left side is $f(x^3-y) = y-x^3$ and the right side is $x \cdot (-x^2)+y = y-x^3$.

Write $P(x, y)$ for the equation. $P(0, y)$ gives $f(f(y)) = y$, so f is a bijection. Let $c = f(0)$. $P(x, f(t))$ gives $f(x^3+t) = xf(x^2)+f(t)$, and with $t = 0$ this is $f(x^3) = xf(x^2)+c$. Subtracting, $f(x^3+t) = f(x^3)+f(t)-c$ for all x, t. Every real number is a cube, so $g = f-c$ satisfies $g(s+t) = g(s)+g(t)$ for all real s, t. In terms of g, the relation $f(x^3) = xf(x^2)+c$ reads $g(x^3) = xg(x^2)+cx$, and $x = 1$ gives $c = 0$. Hence f is additive (so odd), and $f(x^3) = xf(x^2)$ for all x.

Let $a = f(1)$. Replacing x by $x+1$ in $f(x^3) = xf(x^2)$ and expanding with additivity:

$$f(x^3)+3f(x^2)+3f(x)+a = (x+1)\big(f(x^2)+2f(x)+a\big).$$

Cancelling $f(x^3) = xf(x^2)$ leaves $2f(x^2)+f(x) = 2xf(x)+ax$. Replacing x by $-x$, which changes the sign of $f(x)$ and leaves $f(x^2)$ unchanged, gives $2f(x^2)-f(x) = 2xf(x)-ax$. Subtracting, $f(x) = ax$ for every x. Then $f(f(y)) = a^2y = y$, so $a = \pm 1$, and both functions were checked above.

Problem 5

The solutions are $f(x) = 0$, $f(x) = x$ and $f(x) = -x$. Each works, since $(\pm x)^3 = \pm x^3$, so in each case both sides equal $f(x^3)+f(y)$ with f linear.

Putting $x = 0$ gives $f(0)^3 = 0$, so $f(0) = 0$; putting $y = 0$ gives $f(x^3) = f(x)^3$. So the equation reads $f(x^3+y) = f(x^3)+f(y)$, and since every real number is a cube, f is additive. Hence $f(qx) = qf(x)$ for rational q. Let $a = f(1)$; then $a = f(1^3) = a^3$, so $a \in \{-1, 0, 1\}$.

Fix a real x and let q be rational. Expanding with additivity,

$$f\big((x+q)^3\big) = f(x^3)+3q\,f(x^2)+3q^2f(x)+q^3a, \qquad f(x+q)^3 = \big(f(x)+qa\big)^3 = f(x)^3+3qa\,f(x)^2+3q^2a^2f(x)+q^3a^3.$$

These are equal for every rational q, and both are polynomials in q, so their coefficients agree. The coefficient of q^2 gives $f(x) = a^2f(x)$, and that of q gives $f(x^2) = af(x)^2$.

If $a = 0$, the first relation gives $f(x) = 0$ for all x. If $a = \pm 1$, let $h = af$. Then h is additive and $h(x^2) = a^2f(x)^2 = f(x)^2 \geq 0$, so $h \geq 0$ on $[0, \infty)$. By Theorem 1 (with h bounded below on $(0, 1)$), $h(x) = h(1)x = a^2x = x$, so $f(x) = ax$. This gives $f(x) = x$ and $f(x) = -x$.

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Problem 6

The solutions are $f(x) = 0$, $f(x) = 1$ and $f(x) = x$. Check: for $f(x) = 0$ both sides are 0; for $f(x) = 1$ both sides are 2; for $f(x) = x$ the left side is $x^2 + y + x$ and the right side is $x(x + 1) + y$.

Write $P(x, y)$ for the equation and let $c = f(0)$. Subtracting $P(x, 0)$ from $P(x, y)$ gives $f(x^2 + y) = f(x^2) + f(y) - c$, so $g = f - c$ satisfies $g(t + y) = g(t) + g(y)$ for all $t \geq 0$ and all real y. We show that g is additive. Taking $t = y = 0$ gives $g(0) = 0$. For $t < 0$, the rule applied to $-t > 0$ gives $0 = g(0) = g(-t) + g(t)$ and $g(y) = g\big((y + t) + (-t)\big) = g(y + t) + g(-t)$; together these give $g(y + t) = g(y) + g(t)$. So g is additive, and in particular odd.

Let $b = g(1)$, so that $f(x + 1) = g(x) + b + c$. Substituting $f = g + c$ into $P(x, 0)$, that is $f(x^2) + f(x) = f(x)f(x + 1) + c$, and expanding,

$$g(x^2) = g(x)^2 + (b + 2c - 1)\, g(x) + c(b + c - 1). \qquad (1)$$

Replacing x by $-x$ changes the sign of $g(x)$ and leaves $g(x^2)$ unchanged, so subtracting the two versions of (1) gives $2(b + 2c - 1)g(x) = 0$ for every x.

If g is identically zero, f is the constant c, and the equation becomes $2c = c^2 + c$, so $c = 0$ or $c = 1$.

Otherwise $g(x_0) \neq 0$ for some x_0, so $b + 2c = 1$. Putting $x = 0$ in (1) gives $c(b + c - 1) = 0$, so (1) becomes $g(x^2) = g(x)^2$. Hence $g \geq 0$ on $[0, \infty)$, and by Theorem 1 (with g bounded below on $(0, 1)$), $g(x) = bx$ for all x. Now $x = 1$ in $g(x^2) = g(x)^2$ gives $b = b^2$, and $b \neq 0$ because g is not identically zero, so $b = 1$ and $c = \frac{1-b}{2} = 0$. This gives $f(x) = x$.

Note that the constant solution $f(x) = 1$ is easy to lose: it has $g = 0$, so it appears only in the first case.

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